

PLOIESTI, Romania

WMO: 153770

Lat: 44.9558N

Lon: 25.9875E

Elev: 178

StdP: 99.20

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.5	-10.2	-15.9	1.0	-11.6	-12.4	1.3	-9.0	11.5	0.7	8.1	2.0	1.0	20	0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.5	33.0	20.9	31.6	20.6	30.2	20.4	23.1	30.1	22.2	28.7	21.4	27.9	2.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.9	15.9	26.2	20.1	15.1	25.2	19.2	14.3	24.3	69.3	30.1	65.8	28.8	63.0	27.8	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.8	6.3	5.3	-18.0	35.6	3.8	1.8	-20.7	36.9	-22.9	38.0	-25.1	39.0	-27.8	40.3		
			-18.1	25.0	3.8	1.3	-20.9	26.0	-23.1	26.8	-25.3	27.5	-28.1	28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	-1.2	2.0	5.9	11.4	16.9	21.3	22.5	22.6	18.0	11.6	5.6	0.6
	DBStd	9.27	4.51	4.76	4.22	3.97	3.30	2.83	2.58	2.99	4.00	3.85	4.41	4.61
	HDD10.0	1201	346	226	135	31	2	0	0	0	0	27	143	291
	HDD18.3	2931	604	458	385	208	64	8	2	3	57	209	382	549
	CDD10.0	1745	0	1	8	73	216	339	388	392	239	78	11	0
	CDD18.3	434	0	0	0	1	21	97	132	137	46	2	0	0
	CDH23.3	4351	0	0	1	36	263	926	1247	1386	461	32	0	0
	CDH26.7	1559	0	0	0	4	52	304	463	587	146	3	0	0
Wind	WSAvg	2.0	1.8	2.0	2.5	2.5	2.3	2.1	1.9	1.9	2.0	1.8	1.8	1.8
Precipitation	PrecAvg	663	36	29	37	54	76	90	75	56	59	56	42	48
	PrecMax	937	76	85	81	160	174	174	169	223	122	138	99	116
	PrecMin	359	4	2	1	17	14	49	24	19	12	2	3	0
	PrecStd	141	20	21	23	36	38	32	38	41	33	32	28	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	18.6	22.3	27.0	30.5	33.2	34.2	35.0	32.8	26.5	20.6	14.0
		MCWB	6.9	11.6	12.3	16.2	18.7	22.0	21.9	21.2	19.9	17.3	14.2	9.6
	2%	DB	8.7	14.7	18.5	23.8	28.0	31.4	32.5	33.2	30.2	23.5	17.3	11.0
		MCWB	6.0	9.4	10.8	15.0	18.3	21.3	21.4	20.4	18.5	15.8	12.4	8.7
	5%	DB	6.3	11.5	16.1	21.4	26.3	29.9	30.9	31.8	28.1	21.1	14.3	8.6
		MCWB	4.8	7.8	9.5	13.5	17.3	20.5	21.2	20.2	18.6	14.8	10.8	6.4
	10%	DB	4.5	8.7	13.7	19.4	24.4	28.4	29.4	30.1	25.9	18.6	11.8	6.4
		MCWB	3.3	5.9	8.2	12.8	16.3	19.9	20.8	20.3	17.8	13.7	9.4	4.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.0	12.5	13.0	17.0	20.8	23.7	24.5	24.5	21.9	18.5	14.5	10.7
		MCDB	10.1	17.3	20.5	25.0	28.3	30.6	30.6	31.6	29.1	25.2	19.8	12.5
	2%	WB	6.4	9.9	11.4	15.7	19.4	22.4	23.1	22.9	20.6	16.5	13.0	8.6
		MCDB	7.9	14.3	17.9	22.8	26.2	29.5	29.8	30.1	27.5	21.8	16.8	11.2
	5%	WB	5.0	8.0	10.0	14.5	18.2	21.5	22.2	21.8	19.2	15.3	11.6	6.7
		MCDB	6.4	11.1	15.0	20.3	24.4	28.2	28.6	28.2	25.3	20.1	14.0	8.6
	10%	WB	3.5	6.4	8.8	13.4	17.0	20.7	21.4	21.0	18.2	14.2	10.0	5.2
		MCDB	4.5	8.4	12.7	18.3	23.0	27.0	27.5	27.6	24.5	17.8	11.6	6.4
Mean Daily Temperature Range	5% DB	MDBR	8.1	9.3	10.6	11.3	11.8	11.8	11.9	12.5	12.0	10.6	8.0	8.1
		MCDBR	10.6	14.7	16.3	16.3	15.0	14.3	14.3	14.7	15.6	15.3	13.1	11.7
		MCWBR	7.8	9.6	9.3	8.1	6.3	5.7	5.4	5.0	6.1	7.8	8.2	8.3
	5% WB	MCDBR	9.7	13.8	14.4	14.3	13.6	13.2	12.8	13.3	13.7	13.9	11.9	11.0
		MCWBR	7.3	9.3	8.5	7.5	6.6	5.9	6.0	6.1	6.2	7.6	7.6	8.1
Clear-Sky Solar Irradiance	taub	0.334	0.366	0.404	0.459	0.434	0.440	0.452	0.468	0.425	0.392	0.363	0.336	
	taud	2.312	2.224	2.173	2.044	2.191	2.213	2.188	2.132	2.217	2.288	2.318	2.325	
	Ebn at Noon	760	799	818	802	838	833	817	785	788	761	716	714	
	Edh at Noon	81	107	129	159	142	139	141	144	122	99	81	73	
All-Sky Solar Radiation	RadAvg	1.36	2.19	3.36	4.55	5.65	6.09	6.18	5.54	4.03	2.66	1.48	1.13	
	RadStd	0.18	0.30	0.36	0.47	0.46	0.46	0.44	0.36	0.34	0.32	0.17	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (64 neighbors)	+0.51	N/A	N/A	N/A	N/A	N/A	N/A	-150	+73	+35

Nomenclature: See separate page